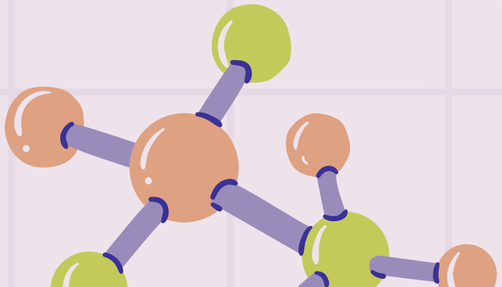




IWSA PROPOSAL

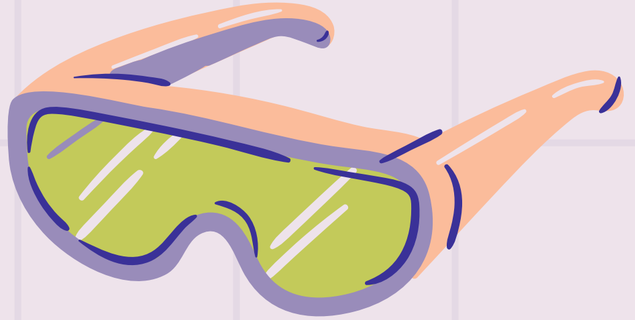
PRESENTED BY : VIDIKA MIRPURI



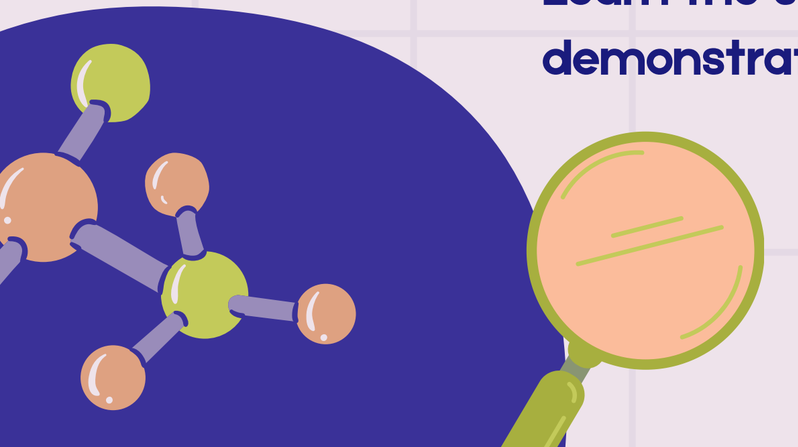




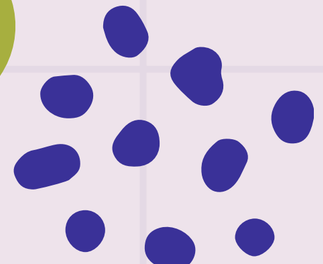
INTRODUCTION



- Aim: Make physics engaging and accessible through low-cost, hands-on toys made from everyday materials.
- Developed 4 working models: Ball Centrifuge, Homopolar Motor, Balloon Hovercraft, Jumping Toy
- Plan to design and develop more innovative physics models covering a wider range of concepts.
- Curate DIY kits containing all the materials needed to build selected toys.
- Conduct interactive workshops in schools where students:
- Build the toys themselves with my guidance.
- Explore physics through hands-on experimentation.
- Learn the scientific principles behind each model through demonstrations and discussions.





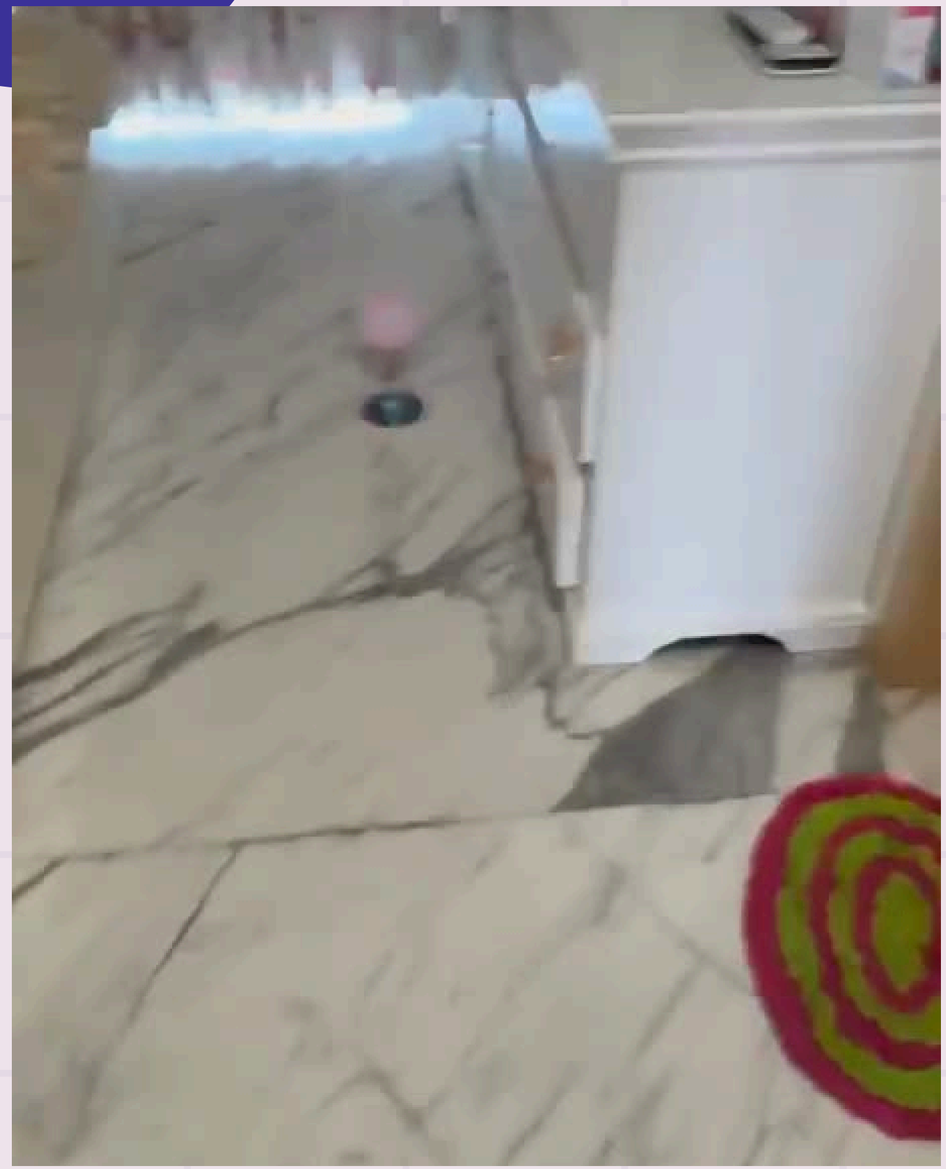


DC MOTOR



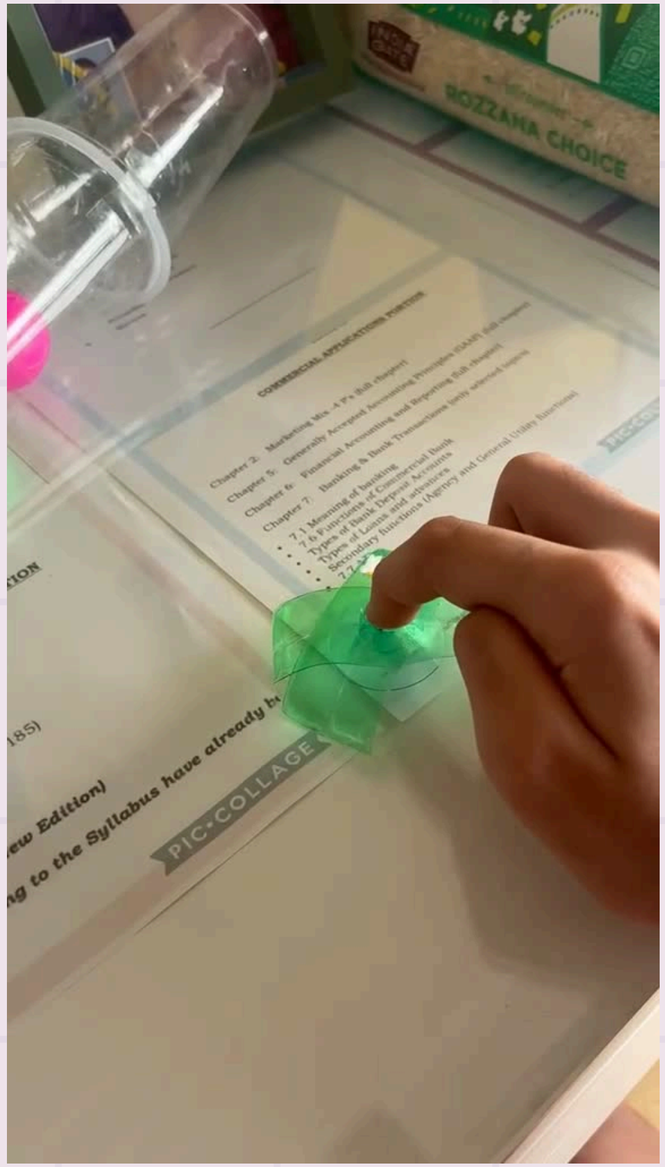
- Children will be amazed to see a wire spin around a battery using just a magnet, and can experiment by changing the magnets to see how the spinning speed changes.
- They will learn how electric current and magnetic fields interact to produce motion, introducing the basic principle behind electric motors.

BALLOON HOVERCRAFT



- Children will love seeing what looks like a floating disc zoom across the floor with almost no effort, and can race different hovercrafts or experiment with balloon sizes to see which travels the farthest.
- They will learn how a cushion of escaping air reduces friction, allowing the hovercraft to glide smoothly while exploring the concepts of air pressure, friction, and motion.

JUMPING TOY



- Children can decorate the toy by drawing and attaching any animal or character they like, making it their own before watching it suddenly leap into the air. They can also experiment with how firmly they press it to see how it affects the jump.
- They will learn how elastic potential energy stored in the suction cup is converted into kinetic energy, causing the toy to spring upwards.

CENTRIFUGE SPINNER



- Children will enjoy spinning the bottle and watching the colourful balls move in unexpected ways. They can experiment by changing the spinning speed to see how the balls behave differently.
- They will learn about centrifugal effects and circular motion, understanding why objects appear to move outward while travelling in a circular path.

SPECTROMETER





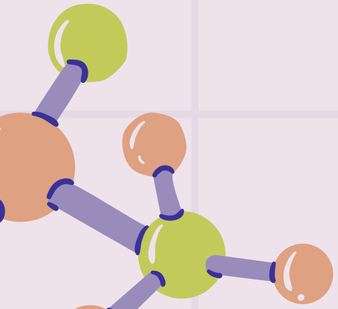
Activity

Children build a simple spectrometer that splits white light into its rainbow spectrum. They compare the spectra of different light sources such as LEDs, fluorescent lamps and sunlight.



What Children Learn

- Every star emits a unique spectrum, like a cosmic fingerprint.
- By studying these spectra, astronomers can determine a star's temperature, chemical composition and age.
- Children can experiment with a real scientific instrument that demonstrates one of astronomy's most powerful techniques for studying the Universe.

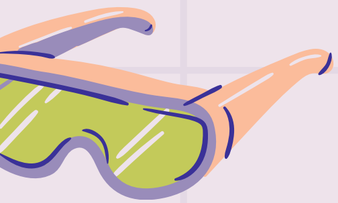


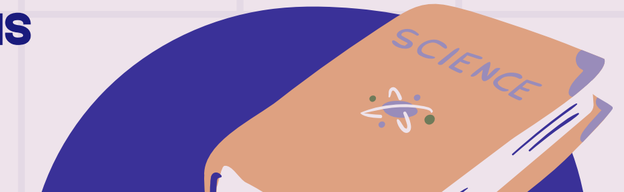
BLACK HOLE DEMO

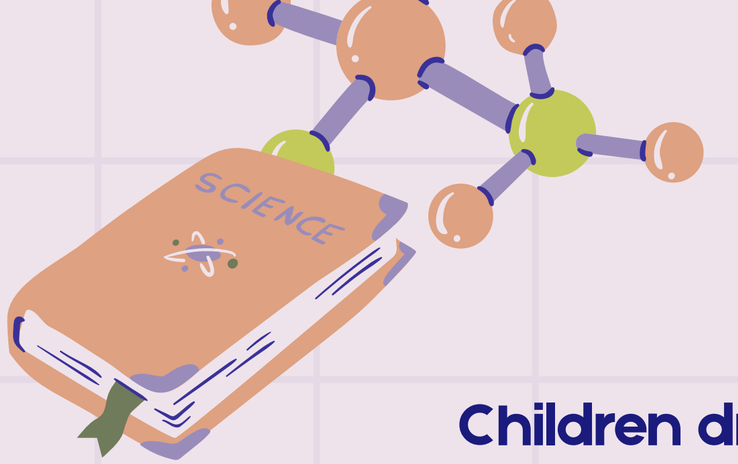
Children stretch the fabric over a frame and place a heavy ball in the centre to represent a massive object. They roll marbles around it, observing how their paths curve and how different speeds produce different orbits.

What Children Learn

- Massive objects such as stars and black holes warp space, causing planets, moons and asteroids to orbit them.
- Gravity governs the motion of celestial objects throughout the Universe, from planetary systems to galaxies.
- The same gravitational interactions influence many of the transient events that astronomers study, such as neutron star mergers and black holes.





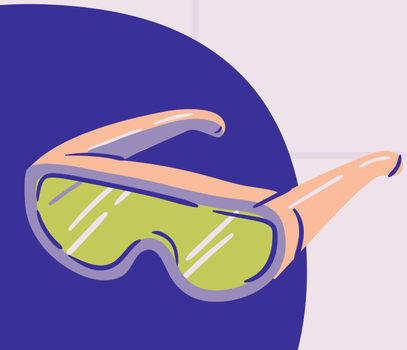


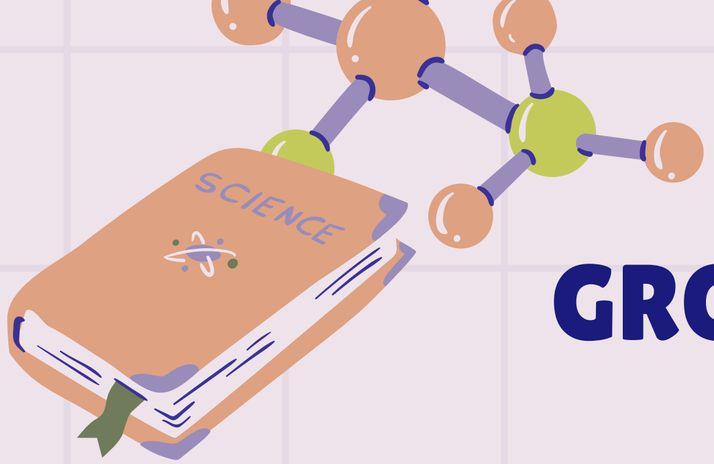
METEORITE IMPACT CRATERS

Children drop different-sized balls from varying heights into the tray and observe the craters formed. They compare the diameter, depth and ejecta patterns produced by each impact.

What Children Learn

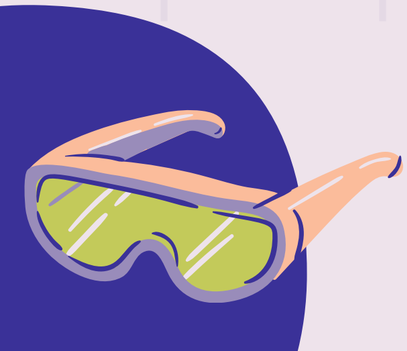
- The size and speed of a meteorite determine the size of the crater it creates.
- Planetary surfaces such as the Moon, Mars and Mercury are covered in impact craters that reveal their history.
- By studying craters, planetary scientists can infer the size, energy and direction of ancient asteroid and meteorite impacts.





GROWTH INDIA TELESCOPE MECHANISMS

- Introduce children to how a modern robotic research telescope works through engaging, low-cost, hands-on demonstrations.
- Rather than only explaining astronomy, each activity recreates one real mechanism used by the GROWTH-India Telescope (GIT), allowing children to discover the concepts themselves.
- Every demonstration will be followed by a discussion connecting the activity to how GIT observes supernovae, asteroids and gamma-ray bursts.

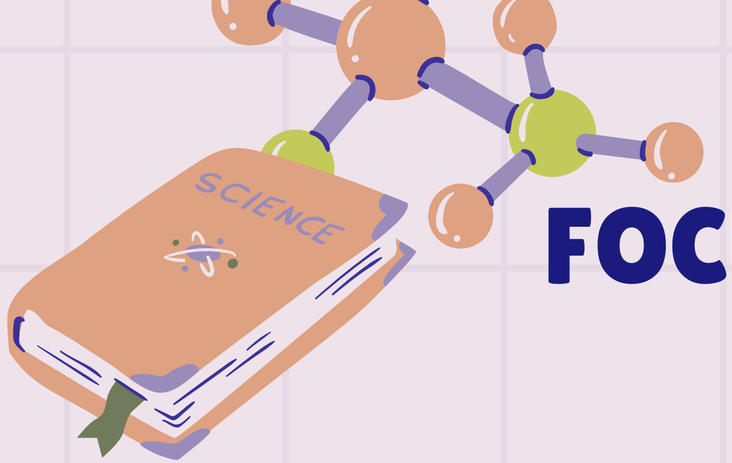




LIGHT COLLECTION (APERTURE)

- Activity: Which Telescope Sees More?
- Children use the two funnels to collect as many "photons" (ping pong balls) as possible within a set time. They compare how many each funnel collects and observe that the wider funnel consistently gathers more.

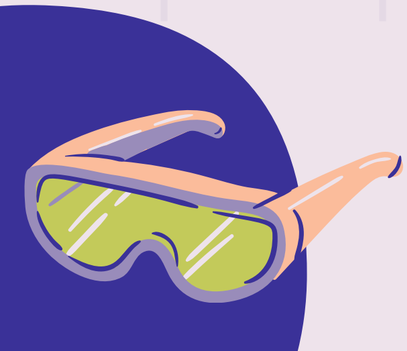
- Demonstrates why GIT's 0.7 m primary mirror collects much more light than a smaller telescope.
- A larger mirror collects more photons, allowing astronomers to observe fainter and more distant objects.

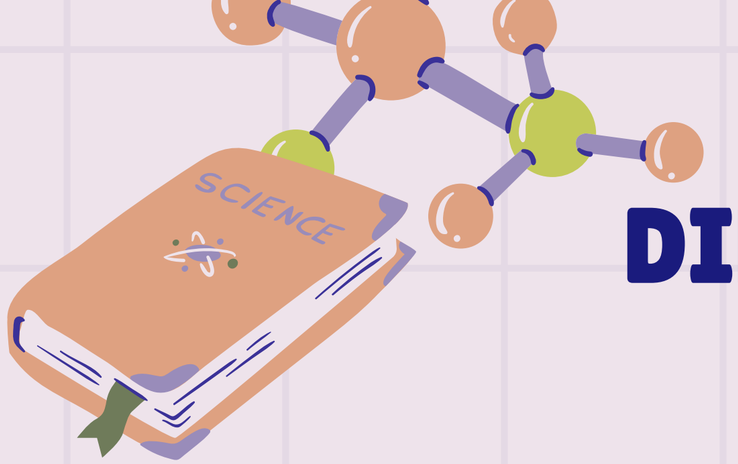


FOCUSING LIGHT (PRIMARY MIRROR)

- Activity: Spoon Mirror Experiment
- Using a torch and the concave side of a metal spoon, children observe how the reflected light changes as the torch is moved. They notice that the curved surface gathers the light into a smaller, brighter spot instead of scattering it.

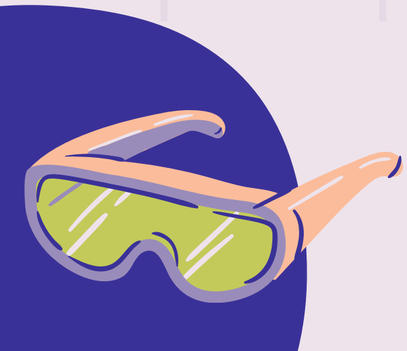
- This demonstrates the role of GIT's concave primary mirror, which collects incoming light from distant astronomical objects and focuses it to a single point. Without this precise focusing, the telescope would be unable to produce the sharp images needed for scientific observations.

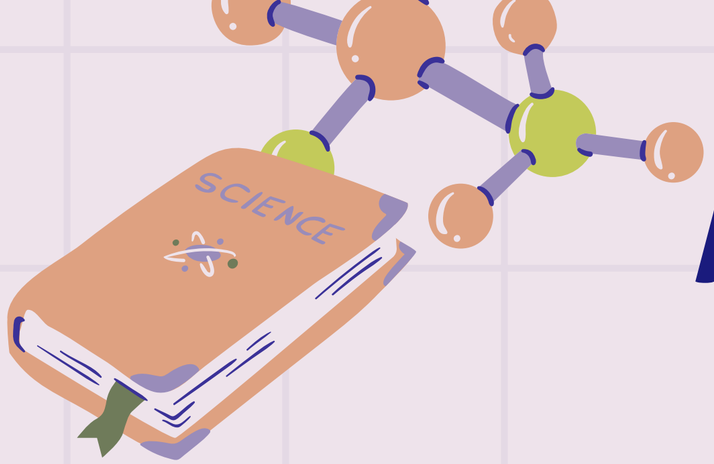




DIRECTING LIGHT TO THE CAMERA

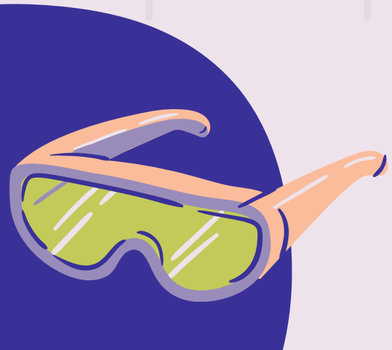
- Activity: Two-Mirror Light Relay
- Children use two hand mirrors to guide a beam of light from a torch onto a target. They experiment with different mirror angles until they successfully redirect the beam through multiple reflections.
- Rather than placing the camera directly in front of the primary mirror, GIT uses a secondary mirror to redirect the focused light through the telescope and onto the CCD detector. This activity helps children understand how mirrors can accurately guide light while preserving the image.

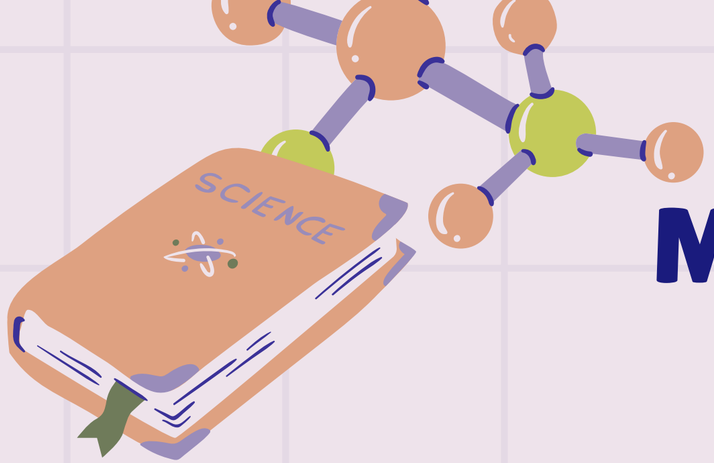




ACCURATE FOCUSING (FOCAL PLANE)

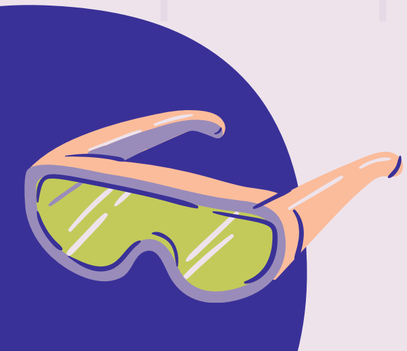
- Activity: Magnifying Glass Focus
- Children project the image of a distant object onto a sheet of paper using a magnifying glass. By moving the paper back and forth, they discover that there is only one position where the image becomes perfectly sharp.
- This illustrates the concept of the focal plane. Inside GIT, the CCD camera must be positioned at this exact location to record sharp images of stars and galaxies. Even a small change in position causes the image to become blurred, highlighting the importance of precise optical alignment.

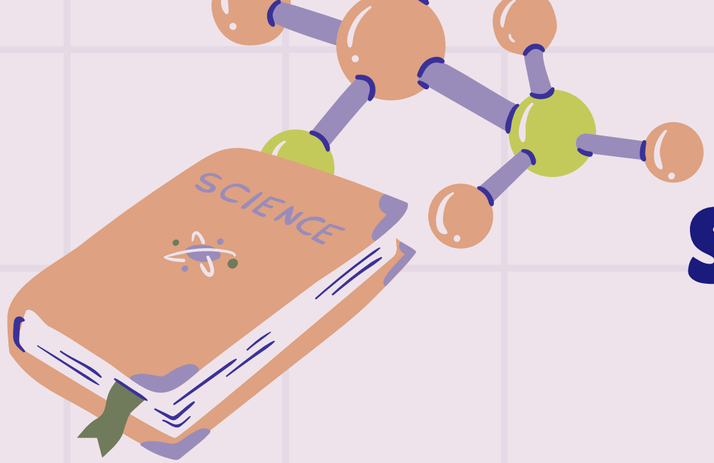




MUTI-FILTER IMAGING

- Activity: Colour Filter Investigation
- Children place different coloured transparent sheets (red, blue and green) over a torch and compare how each filter changes the colour and brightness of the light reaching an image of a galaxy. They discuss why each filter reveals different information.
- GIT captures multiple images of the same object using specialised g′, r′ and i′ filters, each allowing only certain wavelengths of light to reach the detector. By comparing these images, astronomers can determine important properties of celestial objects, including their temperature, composition and how they change over time.

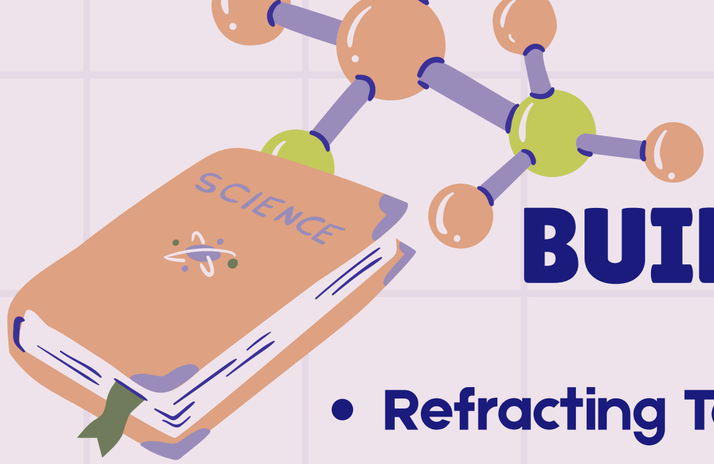




SPOT THE SUPERNOVA

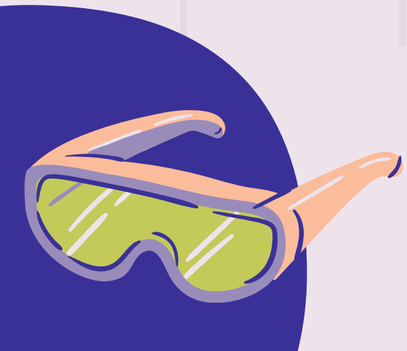
- Activity: spot the supernova
- Children compare two images of the same star field taken on different nights and race to identify the new star that has appeared. The activity uses real astronomical images captured by the GROWTH-India Telescope, obtained during my internship at IIT Bombay.
- Demonstrates how astronomers discover supernovae, asteroids and other transient events by comparing images taken at different times.
- Introduces the image comparison techniques used by GIT to identify newly appearing or changing objects in the night sky.

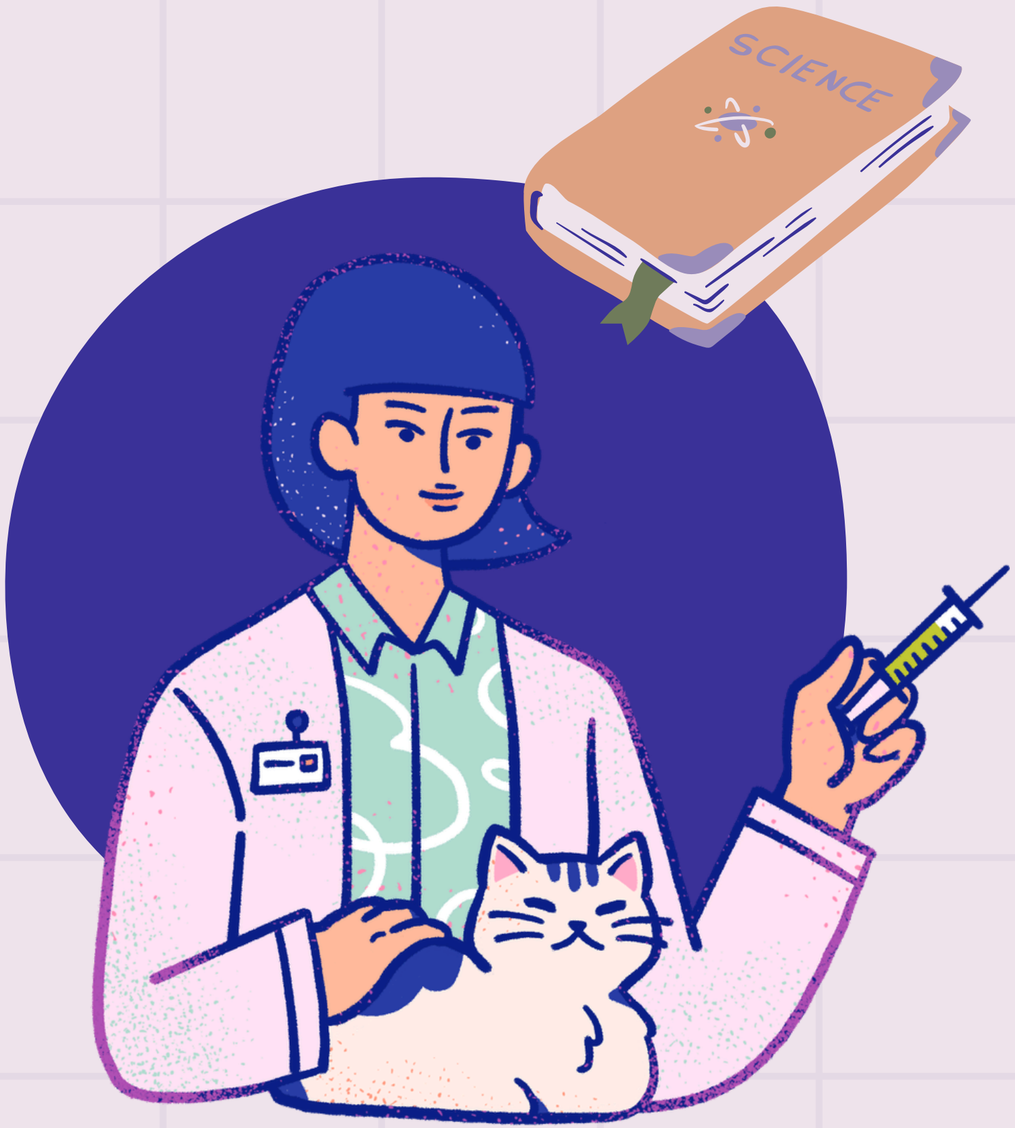


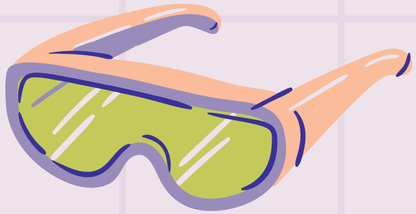


BUILD YOUR OWN TELESCOPE

- Refracting Telescope
- Children conclude the workshop by constructing a simple working refracting telescope using two lenses and sliding cardboard tubes. They then adjust the position of the lenses until distant objects appear in focus, reinforcing the concepts explored throughout the workshop.
- Connection to GIT
- Although GIT is a reflecting telescope, this activity brings together the same fundamental principles explored in the previous demonstrations—collecting light, focusing it accurately and producing a clear image. It allows children to apply these concepts by building and using their own optical instrument.







THANK YOU